

Nguyễn Hoàng Phương Anh

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github.com/phuonganh284 | phuonganh284.github.io/phuonganh-portfolio

EDUCATION

Ho Chi Minh City University of Technology (HCMUT)

Bachelor of Computer Science

Sep 2023 - Present

Ho Chi Minh City, Vietnam

- GPA: 3.1 / 4.0 (7.68 / 10)
- Highest semester GPA: 3.9 / 4.0
- Relevant Coursework: Operating Systems, Data Structures and Algorithms, Computer Networks, Software Engineering, Introduction to Artificial Intelligence

PROJECTS

Smart Home IoT System | IoT Machine Learning FullStack project

Apr 2026

- Role:** Leader - Backend & IoT Developer
- Built backend services for smart home system using Express.js (MVC) and PostgreSQL
- Developed an IoT gateway integrating ESP32 via MQTT and REST APIs
- Implemented JWT authentication and telemetry database design
- Applied ML models on 100K+ records (MAE: 3.34) to predict device behavior and trigger automated control
- Performed load testing with Autocannon: 193 req/sec under 80 concurrent users with 0% error rate

Tech: Express.js, Python, FastAPI, ESP32, PostgreSQL, Supabase

GitHub: https://github.com/phuonganh284/SmartHome_IOT

Library Management System | FullStack project

Dec 2025

- Role:** FullStack Developer
- Developed a FullStack Library management system using Express.js (MVC), React, and PostgreSQL
- Built RESTful APIs for book inventory and user management
- Designed optimized relational database schema
- Implemented responsive UI with React, Vite, Tailwind CSS
- Performed load testing with Autocannon: 73 req/sec with 50 concurrent users

Tech: Express.js, REST API, React, Vite, Tailwind CSS, JavaScript, PostgreSQL, Supabase

GitHub: <https://github.com/phuonganh284/Smart-Library>

Chess AI agent | Searching AI project

Apr 2026

- Role:** Leader - Game Engine & AI developer
- Developed a chess engine with GUI using Python and Tkinter
- Implemented Minimax algorithm with alpha-beta pruning for move decision making
- Designed heuristic evaluation function for board state scoring

Tech: Python, Tkinter, Minimax Alpha-Beta pruning

GitHub: https://github.com/phuonganh284/Chess_AI

SKILLS

Programming Languages: Python, C/C++, C#, JavaScript (ES6+), SQL

Backend: Node.js, Express.js, FastAPIs, REST APIs

Frontend: HTML/CSS, React, Tailwind CSS

Database: PostgreSQL, Microsoft SQL Server, Supabase

Tools & Testing: Git/GitHub, Docker, VS Code, Postman, Unity

CERTIFICATIONS

IELTS Academic: 7.5 Overall (2024)

Microsoft Office Specialist (MOS): Excel & Word